Applied Data Science (STAT GR 5243) Project 4

Prediction on Mortgage Loan Application

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Github Repository: https://github.com/xsxsx-999/5243-project4

Mortgage Approval Estimator ShinyApp: https://crystalguo.shinyapps.io/mortgage-approval-estimator/

1. Project Overview

1.1 Background and Objective

The objective of this project is to build an end-to-end machine learning pipeline to analyze mortgage loan decisions within the United States mortgage market. Using data from the Home Mortgage Disclosure Act (HMDA), this study focuses on applications with definitive outcomes and models whether an application is originated or denied.

By engineering robust features and training predictive models, this project seeks to identify the key drivers—ranging from applicant demographics to loan characteristics—that influence mortgage origination outcomes. The final goal is not only to predict application outcomes, but also to explain the main factors behind model decisions through feature-signal analysis, model interpretation, and fairness-oriented diagnostics.

1.2 Data Collection

To construct a focused and computationally viable dataset, our data acquisition and sampling strategy followed a strict, predefined methodology:

- **Data Source & Geographical Scope:** We obtained historical mortgage application records from the HMDA database through the FFIEC/CFPB HMDA Data Browser API. The dataset spans a six-year period from **2019 to 2024**. To capture large and economically important mortgage markets while managing data volume, the geographical scope was restricted to **California (CA) and Texas (TX)**, two of the largest U.S. housing markets by housing-unit count (FFIEC, 2025).
- **Target Formulation & Filtering:** The full HMDA dataset contains multiple stages of the application lifecycle, including originated, denied, withdrawn, incomplete, and approved-but-not-accepted applications. For the purpose of binary classification, we isolated records to strictly two definitive outcomes. **Originated** is mapped to `target_y = 1`, while **Denied** is mapped to `target_y = 0`.
- **Stratified Sampling:** Given the massive scale of the raw dataset, we applied a **1% stratified proportional sampling fraction** strictly within the restricted two-class subset, preserving the class distribution by design. This yielded a final analytical sample of **139,991** records.

1.3 Target Definition and Statistical Interpretation

Because the analysis filters HMDA records to two definitive outcomes, the model estimates the conditional probability of origination within the Originated-or-Denied subset:

$$P(\text{Originated} \mid \text{Originated or Denied})$$

It does not estimate the unconditional probability of origination across all HMDA applications. Withdrawn, incomplete, approved-but-not-accepted, and other non-final outcomes are excluded from the binary target construction. Therefore, all subsequent model performance, feature importance, threshold analysis, and fairness interpretation should be read as applying only to applications with a definitive Originated-or-Denied outcome.

2. Data Cleaning and Preprocessing

To ensure data integrity and prevent data leakage, the preprocessing pipeline computes all data-driven transformations strictly on the training set, then applies the learned parameters to the test set. This design keeps the feature engineering, imputation, outlier handling, transformation, and encoding steps consistent with the later modeling workflow.

2.1 Datatype Formatting and Feature Engineering

To prevent data leakage, we first removed post-application variables that encode or directly follow from the application outcome. These include the action and denial-reason fields (`action_taken`, `denial_reason-1` through `denial_reason-4`, and `purchaser_type`), as well as loan pricing fields that are recorded only for originated loans (`interest_rate`, `rate_spread`, `total_loan_costs`, `origination_charges`, `discount_points`, `lender_credits`, `hoepa_status`, and `total_points_and_fees`). We then enforced strict numeric and string type casting while preserving missingness for later imputation.

Two additional features were engineered to improve modeling structure. First, `activity_year` was transformed into a categorical `covid_phase` variable to capture macroeconomic shifts across the 2019–2024 study period. Second, HMDA's categorical debt-to-income strings were parsed into an ordinal numeric feature (`debt_to_income_ratio_ord`) together with a binary missingness indicator (`dti_missing_flag`).

2.2 Column Dropping Rules

To reduce noise, prevent unnecessary dimensionality growth, and improve computational efficiency, features were systematically dropped based on missingness, variance, and cardinality thresholds.

Any column with a missing value rate exceeding **80%** was dropped directly, since imputing such heavily sparse features would introduce substantial artificial signal. Features lacking sufficient variation were also removed because they provide little predictive value. For continuous columns, we dropped variables with standard deviation `<= 0.03`. For categorical columns, features were dropped if a single dominant category accounted for `>= 98%` of the observations.

To prevent sparse matrix explosion during one-hot encoding, a tiered cardinality filter was applied to categorical features. Columns with more than **8 unique values** were dropped to minimize noise and computational burden, while columns with **5–8 unique values** were flagged for manual consolidation, such as bucketing rare categories to improve robustness.

2.3 Imputation

Following missingness analysis via heatmap, we applied a tiered imputation strategy. Numeric fields were imputed using `KNNImputer(k=5)` with temporary scaling to support distance-based imputation accuracy, while categorical features received mode imputation.

To maintain interpretability, selected discrete and ordinal features, including `loan_term` and `debt_to_income_ratio_ord`, were explicitly overridden with median values rather than relying only on KNN-based imputation.

2.4 Log1p Transformation

To stabilize variance and mitigate right-skewness, we applied a `log1p` transformation after clipping negative values to zero. This transformation was applied unconditionally to heavy-tailed monetary features, including `loan_amount` and `property_value`.

For `income`, the transformation was applied dynamically only when its training distribution showed strong right skewness, defined as a skewness coefficient greater than `1.0`. This avoids unnecessary transformation when the feature distribution does not require it.

2.5 Outlier Handling

To mitigate the impact of extreme anomalies, we applied Winsorization by clipping numeric values at the 1st and 99th percentiles. The clipping boundaries were derived strictly from the training set and then applied to both training and test splits.

This process was restricted to high-cardinality numeric features with more than 10 unique values, preserving the integrity of binary and ordinal indicators. DTI-related features were explicitly excluded from Winsorization to avoid distorting their ordinal meaning.

2.6 One-Hot Encoding and Final Matrix Construction

We used a `ColumnTransformer` to convert the cleaned feature matrix into a modeling-ready design matrix. Key steps included binarizing `state_code`, treating low-cardinality numeric-coded fields as categorical, and applying `OneHotEncoder` to categorical features. The encoder was configured to handle unseen categories during test-time transformation and to avoid unnecessary collinearity.

After cleaning and imputation, the training matrix used for feature-signal analysis contains **111,992 rows** and **45 predictors**, including **32 numeric** and **13 categorical** features. After one-hot encoding and final matrix construction, the modeling-ready sparse design matrix contains **97 encoded features**.

This final matrix is used as the common handoff point for the downstream EDA, dimensionality-reduction analysis, and supervised modeling stages.

3. EDA: Feature Signal and Redundancy Analysis

This section evaluates which features drive loan approval outcomes. To prevent data leakage, all statistics are computed strictly on the finalized training artifacts (`X_train_imputed` and `y_train`) to ensure consistency with the modeling workflow.

- **Train Size:** 111,992 rows
- **Predictors Analyzed:** 45 (32 numeric, 13 categorical)
- **Numeric Signal:** Point-biserial correlation (with Spearman correlation as a robustness check)
- **Categorical Signal:** Cramer's V
- **Post-OHE Signal:** Approval-rate lift (relative to the base training approval rate)
- **Redundancy Diagnostic:** Spearman correlation (|rho| ≥ 0.8)

3.1 Key Signal Features

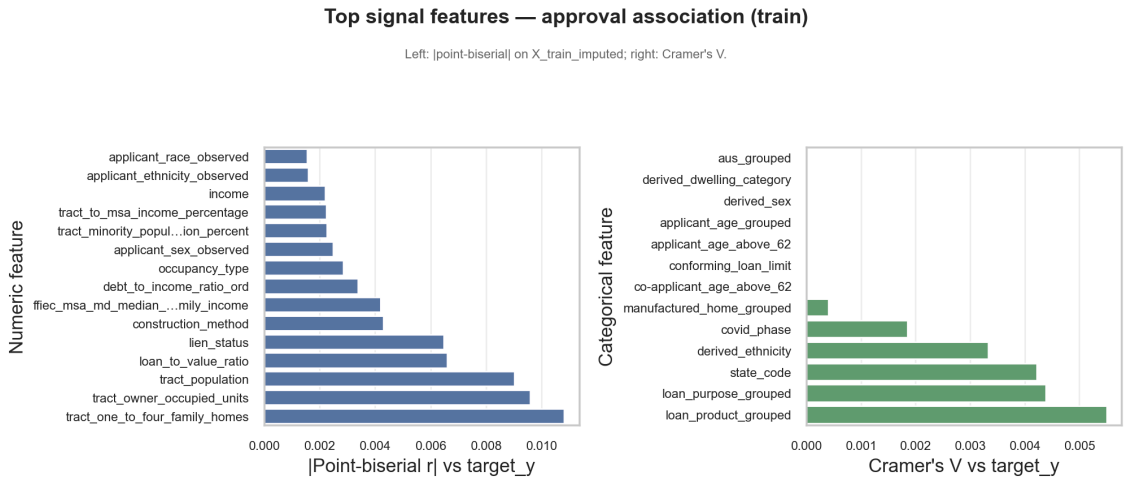


Figure 3.1. Top numeric and categorical features ranked by their association with the target variable on the training split.

The numeric feature rankings are dominated by tract-level demographic and housing statistics (e.g., `tract_one_to_four_family_homes`, `tract_owner_occupied_units`, `tract_population`), alongside loan-specific metrics like `loan_to_value_ratio` and `lien_status`. While individual continuous variables exhibit relatively weak marginal associations with the target, their signals cluster logically around local geography, collateral, and mortgage structure. This supports our downstream use of multivariate models capable of capturing nonlinear combinations and interactions.

For categorical features, the strongest overall associations (measured by Cramer's V) stem from loan products, loan purposes, and applicant demographics. Though absolute correlation values are modest, these fields yield significant predictive power once one-hot encoded.

3.2 Post-OHE Category Lift

To better interpret categorical signals post-encoding, we calculate the "lift" for each One-Hot Encoded (OHE) category. Lift measures the difference between the approval rate when a specific category is active versus the overall base approval rate. We filter for categories with at least 1% support to exclude noisy, low-sample outliers.

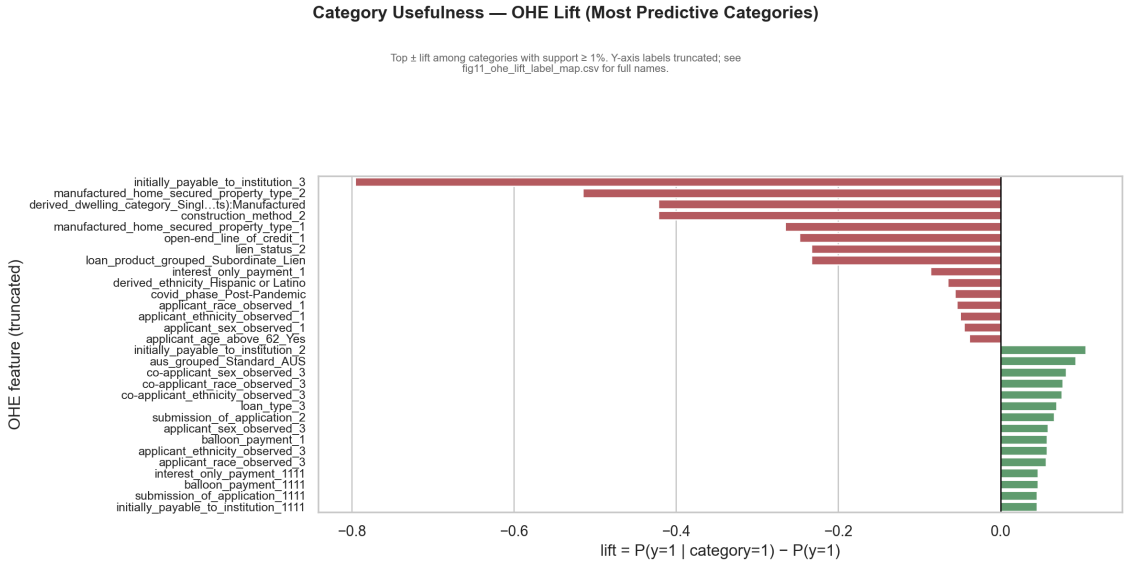


Figure 3.2. Category-level lift relative to the base approval rate, highlighting the strongest positive and negative OHE effects (minimum 1% support).

The strongest **positive-lift** drivers include applications processed via the standard automated underwriting system (`aus_grouped_Standard_AUS`), VA loans (`loan_type_3`), and joint applications. These categories show approval rates roughly 6 to 9 percentage points higher than the baseline. Conversely, the strongest **negative-lift** drivers cluster around specific loan purposes, observation statuses, and demographic flags historically correlated with lower approval odds. Ultimately, while Cramer's V highlights broadly useful categorical fields, lift isolates the specific encoded levels that push predictions positively or negatively.

3.3 Multicollinearity and Redundancy Findings

We identified 11 highly correlated numeric feature pairs (|Spearman| ≥ 0.8). These redundancies primarily surface within related groupings: co-applicant variables, applicant observation variables, tract-level housing counts, and mortgage structure flags.

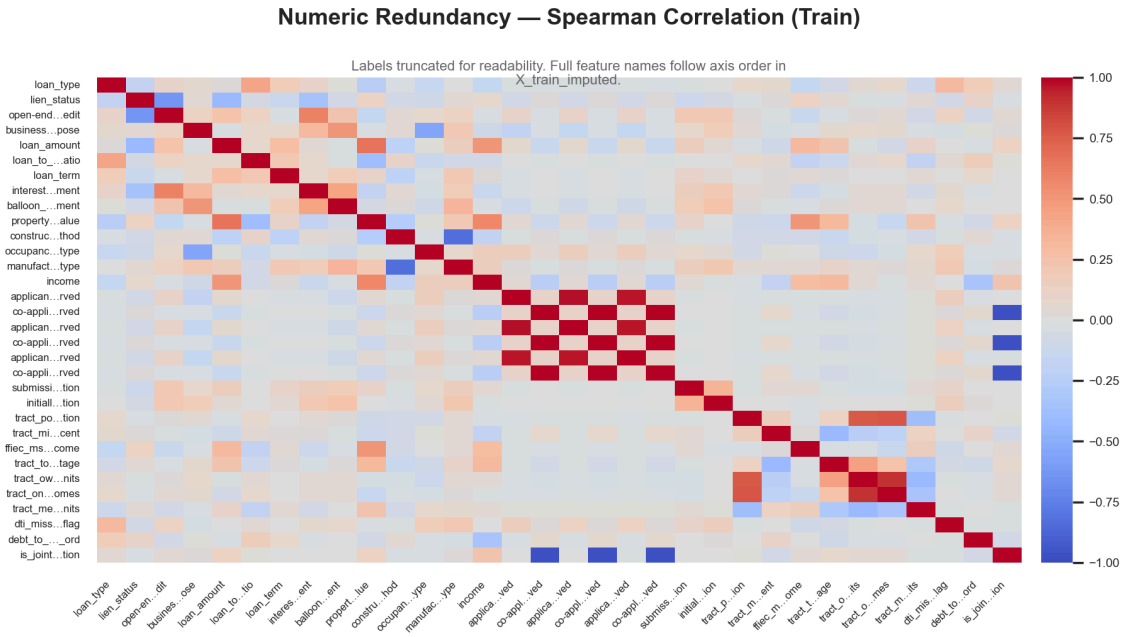


Figure 3.3. Spearman correlation structure among numeric training predictors; dense warm/cool clusters indicate redundant feature groups.

This indicates meaningful redundancy in the raw feature space. While tree-based ensembles handle multicollinearity well, feature redundancy complicates interpretability, as correlated variables can split or dilute attribution. Therefore, we maintain feature-level transparency in our modeling stages rather than treating dimensionality reduction as a blanket solution.

Summary: The EDA confirms that the dataset contains meaningful—albeit weak—supervised signals concentrated in tract structures, loan terms, and specific OHE levels. It also reveals inherent redundancies, motivating the TruncatedSVD diagnostic in Section 4 to understand the dataset's underlying dimensional structure.

4. Unsupervised Learning (TruncatedSVD)

4.1 Method and Controls

Because our finalized preprocessing pipeline outputs a sparse design matrix, `TruncatedSVD` is the optimal dimensionality-reduction method for this analysis. Unlike dense PCA, SVD operates directly on sparse matrices, preserving computational efficiency and avoiding the memory risks of densification.

To prevent data leakage, this unsupervised step is fit exclusively on the training data, with explicit alignment checks against `y_train`. For interpretability, the resulting component loadings are mapped back to their respective encoded feature names.

4.2 Results and Practical Use

We evaluated SVD components based on cumulative explained variance thresholds (90% and 95%). As shown in Figure 4.1, the feature space is highly compressible: the combined sparse input captures 90% of the variance with just 22 latent components, and 95% with 30 components. When evaluating the feature blocks independently, the binary OHE block reaches these thresholds at 21 and 27 components, while the standardized continuous block requires only 8 and 10 components.

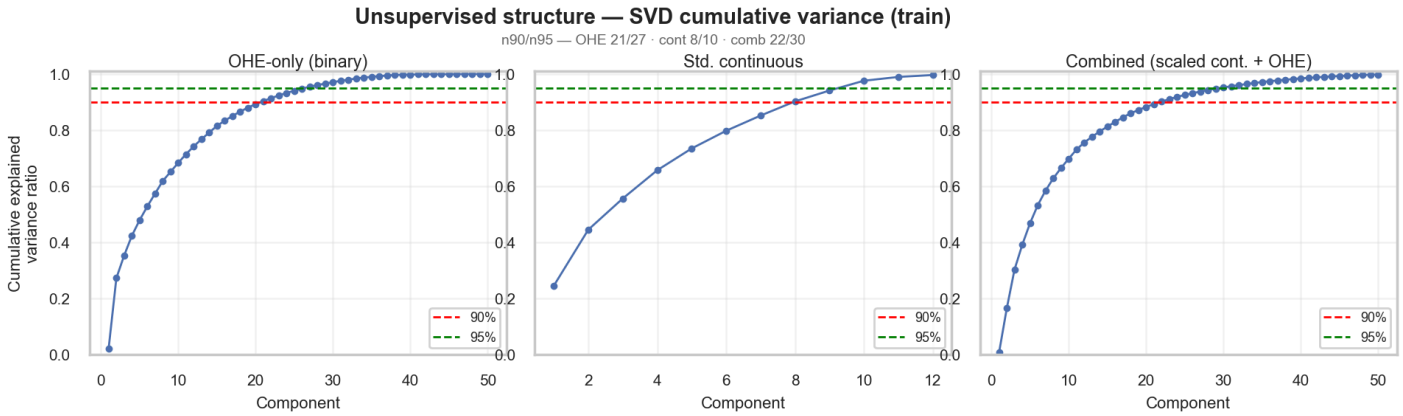


Figure 4.(a). Cumulative explained variance using TruncatedSVD, comparing three feature blocks: binary OHE columns (left), standardized continuous columns (center), and the combined sparse input (right).

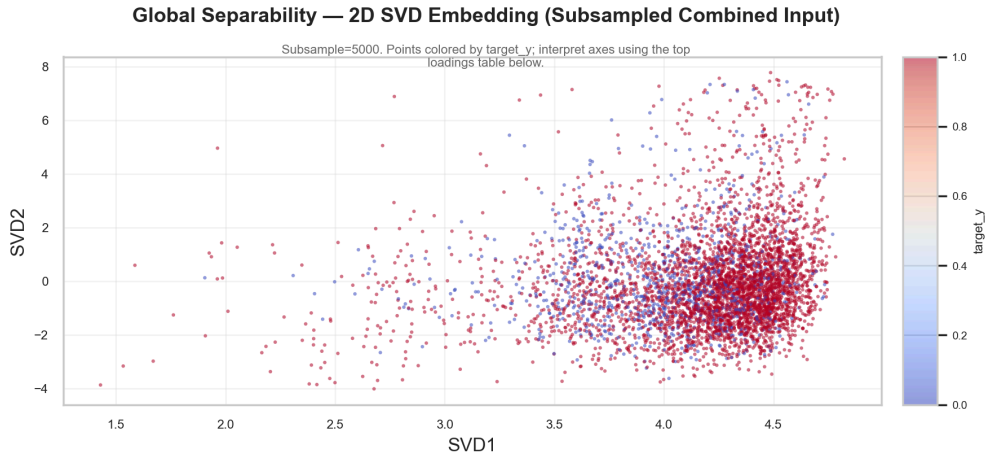


Figure 4.(b). Two-dimensional SVD embedding (subsample) colored post-hoc by `target_y` to visualize global separability. Note: Target labels were not used during SVD fitting.

The 2D embedding (Figure 4.2) serves as a diagnostic visualization rather than a final modeling representation. Top-loading summaries help us identify which original encoded variables drive each latent component. For instance, the first SVD component is dominated by loan-structure and binary indicators, while the second is driven primarily by tract-level housing/population variables alongside continuous loan and income measures.

4.3 Modeling Implications

The SVD analysis confirms that the high-dimensional sparse feature space carries significant redundancy. While representing the data with 22–30 latent components is highly efficient, our final modeling stage will retain the raw feature-level OHE matrix.

We make this trade-off to preserve the strict per-feature interpretability required for the SHAP attribution and fairness audits conducted in Section 5. Because tree-based ensembles are inherently robust to feature redundancy, we can comfortably offset a loss in dimensional compactness for a massive gain in interpretability. Ultimately, feature-signal rankings remain our primary analytical anchor, with SVD providing structural validation.

5. Model Building and Evaluation

5.1 Model Selection and Rationale

The modeling strategy is guided by the EDA findings in Sections 3 and 4. Moderate feature-target correlations, nonlinear relationships, correlated numeric predictors, and latent structure in the encoded feature space suggest that denial outcomes are driven by interactions among multiple variables rather than by a single dominant predictor. These patterns support the use of ensemble methods, while Logistic Regression is retained as an interpretable linear baseline.

Three supervised models were evaluated. Logistic Regression provides a transparent baseline suited to the regulatory context of mortgage lending. Random Forest was included as a nonlinear ensemble model that can handle interaction-heavy and partially redundant features, with class imbalance addressed using `class_weight="balanced"`. LightGBM was included as a boosted-tree model for capturing more complex localized interactions in the imbalanced tabular dataset, with imbalance handled using `scale_pos_weight`.

5.2 Training Setup

All three models are evaluated under 5-fold stratified cross-validation on a training set of 111,992 observations, with stratification ensuring the 20.4% denial rate is preserved across every fold. PR-AUC is used as the primary scoring metric because the target class is imbalanced and the main modeling objective is to identify denial cases reliably. ROC-AUC, precision, recall, and F1 are retained as secondary diagnostic metrics.

5.3 Hyperparameter Tuning

Logistic Regression was tuned using GridSearchCV over penalty type and regularization strength. Random Forest and LightGBM were tuned using RandomizedSearchCV with 20 and 30 iterations respectively, given their larger search spaces. Search ranges and selected values are summarized in Table 5.1.

Model	Parameter	Search Range	Best Value
Logistic Regression	`penalty`	L1, L2	L2
Logistic Regression	`C`	0.01, 0.1, 1, 10	1
Random Forest	`n_estimators`	300, 500	300
Random Forest	`max_depth`	None, 10, 20	20
Random Forest	`min_samples_leaf`	1, 5, 20	5

Model	Parameter	Search Range	Best Value
Random Forest	`max_features`	sqrt, 0.3	0.3
LightGBM	`num_leaves`	31, 63, 127	63
LightGBM	`learning_rate`	0.03, 0.05, 0.1	0.03
LightGBM	`n_estimators`	500, 1000	500
LightGBM	`min_child_samples`	20, 100	20
LightGBM	`subsample`	0.8, 1.0	1.0
LightGBM	`colsample_bytree`	0.8, 1.0	1.0

Table 5.1. Hyperparameter search spaces and selected values.

5.4 Probability Calibration

Because the LightGBM output is interpreted as a continuous denial risk score rather than only as a binary classification label, probability calibration was applied to make predicted risks better align with observed denial rates. This is important for downstream threshold selection in Section 5.6.4, where classification cutoffs depend on the reliability of the predicted probabilities.

Isotonic regression calibration was applied to the LightGBM model by `CalibratedClassifierCV` with `cv = 5`. Isotonic regression was preferred over Platt scaling because it imposes only a monotonicity constraint on the mapping from raw scores to calibrated probabilities, allowing it to correct nonlinear calibration distortions typical of gradient-boosted models.

5.5 Leakage Detection and Remediation

During model interpretation, the `initially_payable_to_institution` feature group appeared as an unexpectedly important predictor, with one dummy variable ranking third in permutation importance. Although leakage screening is ideally completed during data cleaning, this feature was not initially removed because the raw data description did not clearly indicate that the variable was unavailable at the time of application. Its unusually high importance prompted a closer review of the feature's timing and meaning.

The review showed that the variable represents the institution to which the loan was initially payable, which is determined after loan origination rather than during the application stage. To avoid target leakage, all four `initially_payable_to_institution` dummy variables were removed, reducing the feature set from 97 to 93 predictors. The calibrated LightGBM model was then refit on the corrected feature set. All subsequent evaluation, interpretation, and threshold analysis are based on this post-removal model.

5.6 Interpretability Methods

Three complementary methods are used to interpret the final LightGBM pipeline. SHAP values are computed with TreeSHAP on a raw LightGBM model refit using the final 93-feature set and selected hyperparameters, supporting both global feature ranking and local case-level explanations. Permutation importance is calculated on the calibrated model as the drop in PR-AUC after each test-set feature is shuffled. Partial Dependence Plots are then generated for the top features to show their marginal relationships with predicted approval probability; denial risk is interpreted in the opposite direction.

5.7 Results

5.7.1 Cross-Validation Performance

Cross-validation results in Table 5.2 show a clear separation between the linear baseline and the two ensemble models during model selection. Logistic Regression achieved a mean ROC-AUC of 0.6911, while Random Forest and LightGBM both exceeded 0.885. This gap supports the EDA finding that mortgage denial outcomes are better captured by nonlinear and interaction-based models than by a purely linear specification. LightGBM achieved the highest mean ROC-AUC and PR-AUC, with the lowest fold-to-fold variance, and was therefore selected as the final model candidate for calibration and refitting.

Model	ROC-AUC	PR-AUC
Logistic Regression	0.6911 ± 0.0039	0.8628 ± 0.0021
Random Forest	0.8854 ± 0.0008	0.9554 ± 0.0009
LightGBM	0.8874 ± 0.0006	0.9565 ± 0.0007

Table 5.2. Cross-validation performance summary during model selection, reported as mean ± standard deviation across 5 folds.

5.7.2 Leakage-Adjusted Test-Set Performance and Calibration

After removing the leakage features, the calibrated LightGBM model was refit on the corrected 93-feature set and evaluated on the held-out test set. The final model achieved a ROC-AUC of 0.8577 and PR-AUC of 0.9455, indicating strong ranking performance after leakage remediation. The Brier score of 0.0952 is used to assess probability calibration, while the denial-class F1 score of 0.6357 summarizes the precision-recall balance for identifying likely denials. These results are treated as the final deployment-relevant performance estimates.

Model	Feature Set	ROC-AUC	PR-AUC	Brier	F1 (Denial)
Calibrated LightGBM	93 features	0.8577	0.9455	0.0952	0.6357

Table 5.3. Leakage-adjusted final model performance on the held-out test set.

5.7.3 Feature Importance and Interpretability

SHAP values show that the final LightGBM model relies most heavily on `debt_to_income_ratio_ord`, `aus_grouped_Standard_AUS`, `loan_to_value_ratio`, `loan_amount`, and `property_value` (Figure 5.1). These predictors are meaningful in a mortgage approval context. Debt-to-income ratio reflects the applicant's repayment burden, while loan-to-value ratio captures how much of the property value is financed by the loan. Their high importance suggests that the model's predictions are driven mainly by affordability and collateral-risk factors. The importance of `aus_grouped_Standard_AUS` also indicates that the automated underwriting system category carries substantial information about the final application outcome.

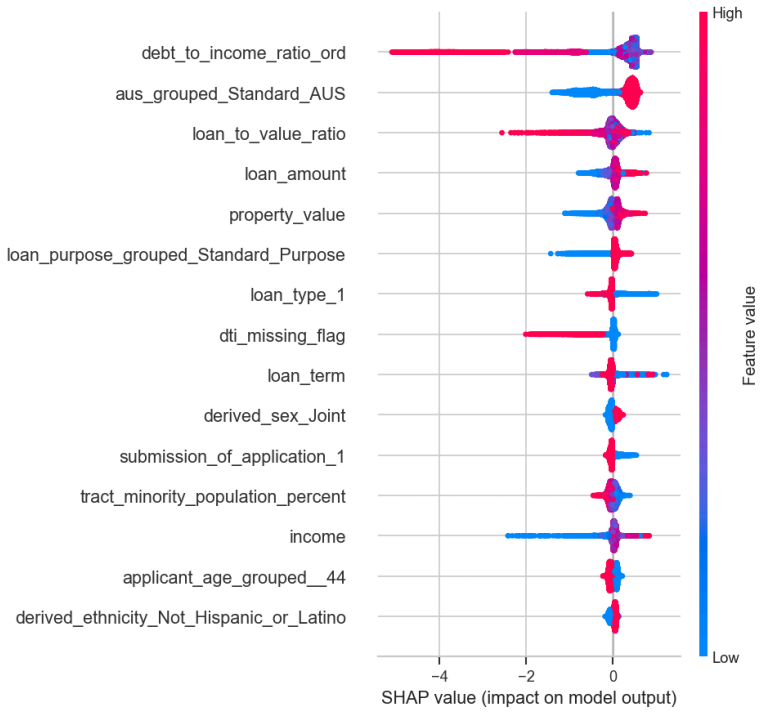


Figure 5.1. SHAP summary plot for the leakage-adjusted LightGBM model.

Partial Dependence Plots further clarify how the top predictors affect predicted approval probability (Figure 5.2). Approval probability drops sharply when `debt_to_income_ratio_ord` reaches high levels, which is consistent with the idea that heavier debt obligations reduce repayment capacity. `loan_to_value_ratio` also shows weaker approval patterns at higher values, reflecting greater risk when the loan amount is large relative to the property value. `loan_amount` and `property_value` have milder marginal effects, suggesting that they provide supporting context rather than acting as dominant decision drivers.

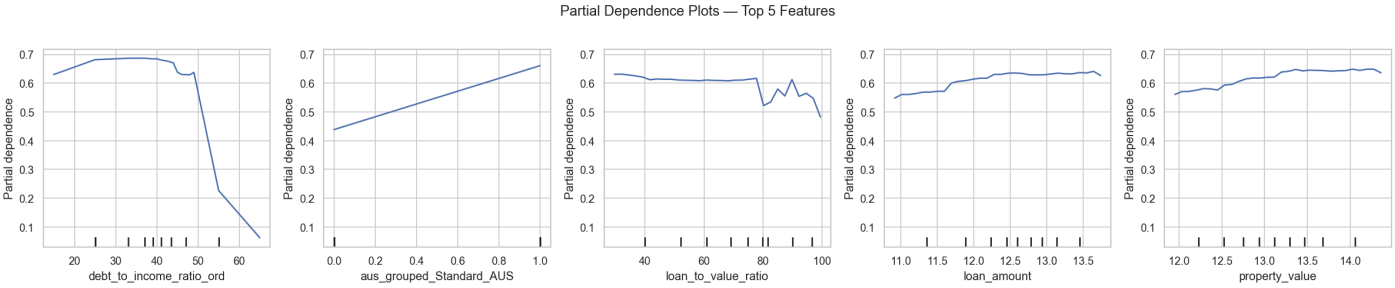


Figure 5.2. Partial dependence plots for the top five features.

5.7.4 Threshold Analysis

Threshold analysis translates the final calibrated model probabilities into an operational decision rule. Because the model predicts approval probability, applications below a chosen threshold are flagged as likely denials for further review. A lower threshold flags only the highest-risk applications, while a higher threshold flags more applications.

Table 5.4 shows the tradeoff between identifying more true denial cases and limiting unnecessary review. As the threshold increases from 0.3 to 0.7, denial recall, or the share of actual denials captured by the model, rises from 0.393 to 0.632. At the same time, the flagged rate, or the share of all applications sent for review, increases from 8.8% to 19.0%.

The denial-class F1 score, which balances precision and recall, is highest at threshold 0.7. At this threshold, the model achieves precision of 0.678, recall of 0.632, and a flagged rate of 19.0%. However, threshold 0.6 produces nearly the same F1 score while flagging only 15.6% of applications. For this reason, 0.6 is a more practical default operating point when the goal is to balance risk detection with review workload.

Threshold	Precision (Denial)	Recall (Denial)	F1 (Denial)	Flagged Rate
0.3	0.911	0.393	0.550	0.088
0.4	0.864	0.457	0.598	0.108
0.5	0.806	0.525	0.636	0.133
0.6	0.753	0.576	0.653	0.156
0.7	0.678	0.632	0.654	0.190

Table 5.4. Threshold sensitivity analysis for the final calibrated LightGBM model.

6. Deployment: Mortgage Approval Estimator

The final calibrated LightGBM model was deployed as an interactive R Shiny application: https://crystalguo.shinyapps.io/mortgage-approval-estimator/. The app integrates the trained model, the isotonic calibration layer, and pre-computed TreeSHAP attributions to support real-time scoring and explanation. All model artifacts are derived from the training and test pipeline described in Sections 2 through 5; the app does not query live HMDA data.

The interface is organized into two tabs. **Tab 1 (Score an Application)** lets users enter applicant details such as income, loan amount, and debt-to-income ratio, and returns an approval probability, a plain-language verdict ("Likely approved", "On the fence", or "Likely denied"), and the top three reasons behind the prediction. **Tab 2 (About the Model)** provides a non-technical overview of how the model was built, its overall accuracy, the most influential features, and its known limitations.

Two extensions are planned for future work: a batch input function that allows users to upload a CSV of applications for portfolio-level scoring, and an exploration dashboard for examining prediction distributions and fairness metrics across demographic and geographic subgroups.

7. Summary and Limitation

This study built an end-to-end machine learning pipeline to predict mortgage application outcomes using HMDA data, with LightGBM selected as the final calibrated model after comparing against Logistic Regression and Random Forest baselines. The model aachieved a test ROC-AUC of 0.858 and PR-AUC of 0.946, with debt-to-income ratio, automated underwriting system, and loan-to-value ratio identified as the strongest predictors of denial.

A primary limitation is that the dataset was restricted to California and Texas to manage data volume. While these are two of the largest U.S. mortgage markets, the model's generalizability to other states with different regional lending patterns, regulatory environments, or demographic compositions has not been tested. Future work could extend the pipeline to additional states and evaluate whether the same feature drivers and threshold tradeoffs hold across more diverse geographic markets.

8. Reference

Federal Financial Institutions Examination Council. (2025). Home Mortgage Disclosure Act (HMDA) Data Browser [Data set]. Consumer Financial Protection Bureau. https://ffiec.cfpb.gov/data-browser/

U.S. Census Bureau. (2025). State Population Totals and Components of Change: 2020–2025 (Vintage 2025) [Data set]. https://www.census.gov/data/tables/time-series/demo/popest/2020s-state-total.html

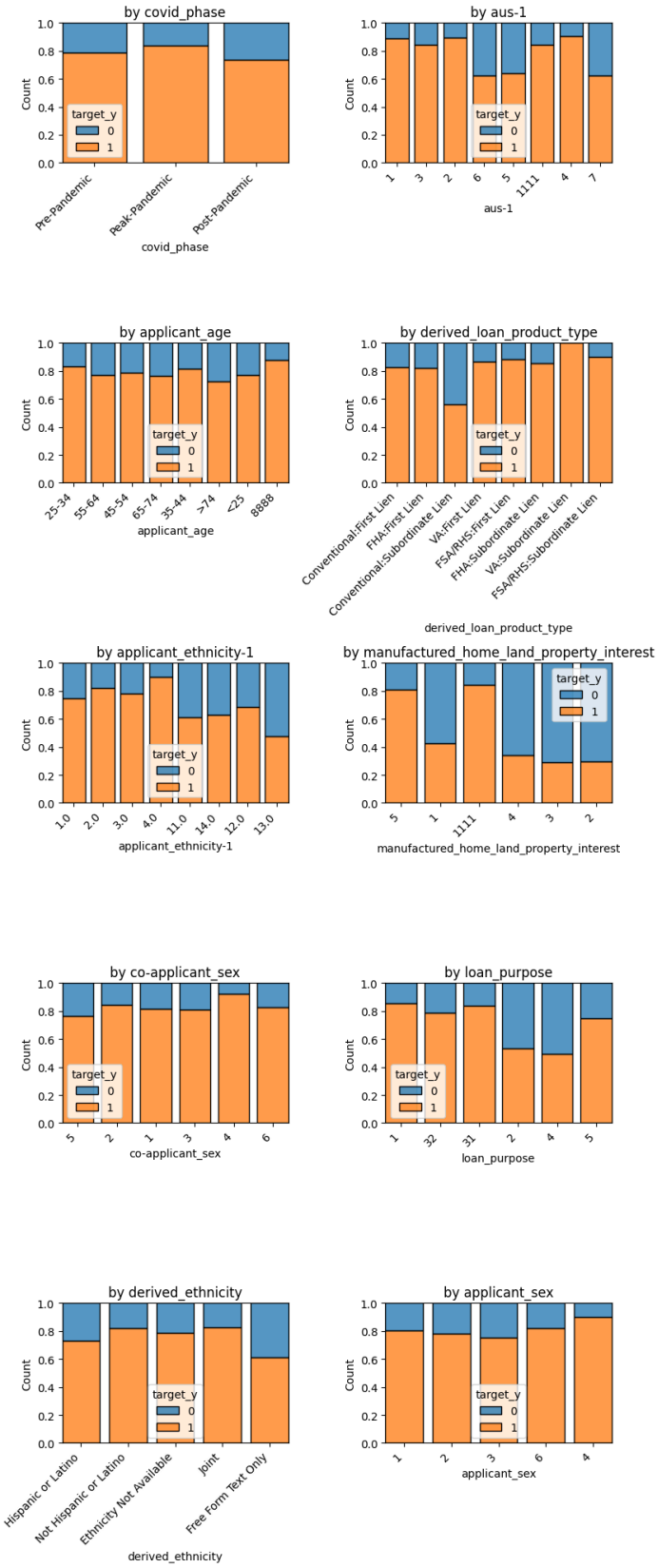
9. Contributions

Team Member	Contribution
Shuxuan Xu (sx2412)	Data preprocessing, including data cleaning and feature engineering. Wrote the corresponding report section.
Fangyi Lin (fl2748)	Exploratory data analysis and visualization. Wrote the corresponding report section.
Veronica Joe (jj3470)	EDA for modeling, including analysis used to support model selection. Wrote the corresponding report section.
Crystal Guo (lg3434)	Model building and evaluation, including model selection and development. Wrote the corresponding report section.

Appendix

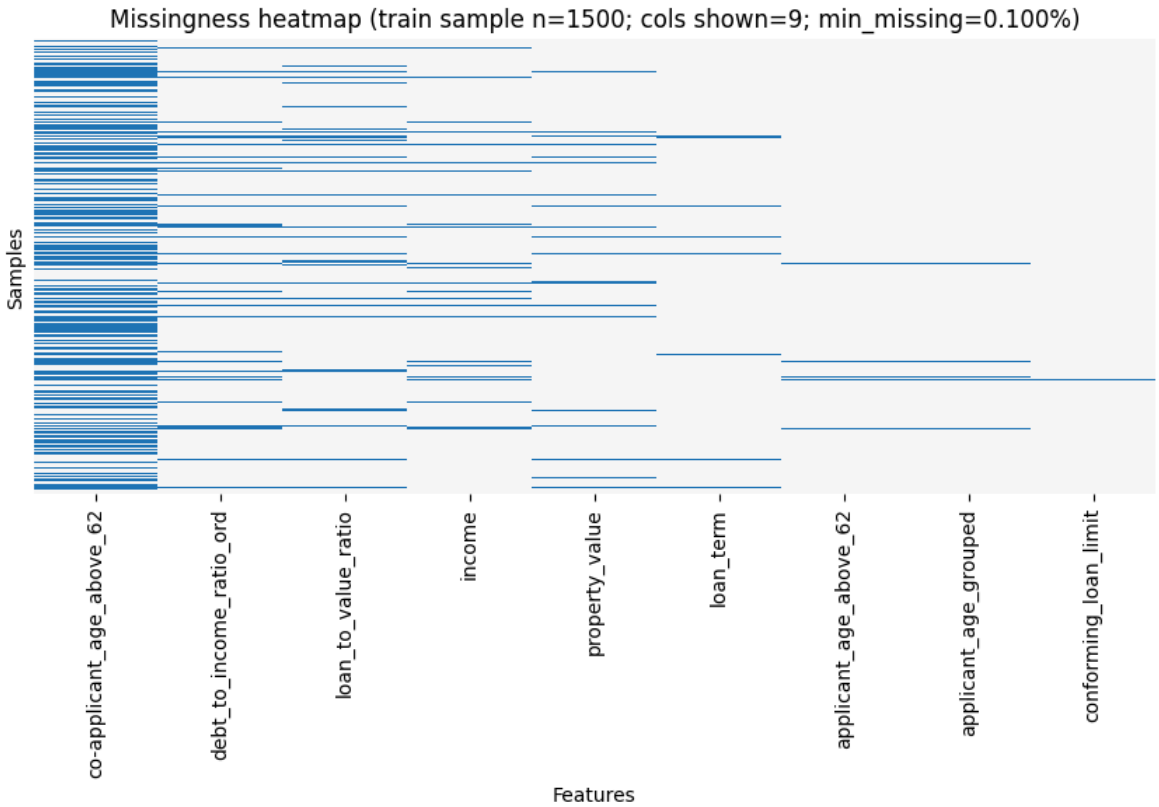
1. Feature Consolidation Details (2.2.3)

Detailed distribution charts



Based on the visual EDA of approval rates across the isolated categorical features, we executed the following consolidation strategy to maximize predictive signal while minimizing dimensionality:

- **Binarized by Presence/Type:** * `co-applicant_sex` : Gender showed zero variance in approval outcomes. The true signal was simply the presence of a co-applicant. Converted to a binary `is_joint_application` (0 = Single, 1 = Joint).
  - `derived_loan_product_type` : Grouped into `First_Lien` (>80% approval) vs. `Subordinate_Lien` (~50% approval).
- **Binned by Risk/Approval Rates:**
  - `aus-1` : Grouped official systems into `Standard_AUS` (codes 1-4, ~90% approval) and mapped the rest to `Non_Standard_or_Exempt` (~60% approval).
  - `loan_purpose` : Grouped standard applications (purchase, refi) into `Standard_Purpose` , and higher-denial applications (home improvement, other) into `High_Risk_Purpose` .
  - `manufactured_home_land_property_interest` : Consolidated into `Has_Property_Interest` and `Not_Applicable_or_Exempt` based on distinct denial rate clusters.
- **Demographic Bucketing:** `applicant_age` was collapsed into broader life-stages (`<25` , `25-44` , `>44`). Anomaly placeholder values (8888, 9999) were explicitly replaced with `NaN` for safe downstream imputation.
- **Dropped:** `applicant_sex` (perfectly flat approval rates across categories; zero predictive value) and `applicant_ethnicity-1` (100% redundant with `derived_ethnicity`).

2. Imputation (2.3)

Heatmap before Imputation